



CYPHER

P2P Audit

Audit Report Completed: August 22, 2025

Contents

1 - Summary	3
1.a - Overview	3
1.b - Process	3
2 - Findings	5
3 - Appendix	5
3.a - Disclaimers	5
3.b - Issue Severity Levels	7
3.c - About Cypher Enterprises	8

1 - Summary

This report is the result of a security audit performed on the P2P cardano-swaps and cardano-loans project(s). The report includes an overview of the project, the process used to conduct the audit, a list of all files audited, and a detailed list of findings. The findings are categorized by severity and include a description of the issue, a recommendation for resolution, and the commit in which the issue was resolved. The report is intended to help the project team identify and resolve any security issues that may exist.

This security assessment represents our best efforts to identify vulnerabilities based on our methodology, but should not be considered exhaustive or definitive. Our focus included examining potential attack vectors that could compromise validator integrity, lead to unauthorized fund manipulation, or result in locked assets. However, due to the inherent complexity of blockchain systems, we cannot guarantee the discovery of all possible vulnerabilities. Security is an evolving process, and additional reviews, ongoing monitoring, and community-driven testing are strongly recommended as complementary security measures.

1.a - Overview

cardano-swaps is a decentralized application built on the Cardano blockchain that enables users to trade Cardano Native Tokens (CNTs). cardano-loans is a decentralized application built on the Cardano blockchain that enables users to borrow and lend Cardano Native Tokens (CNTs).

1.b - Process

In the audit process, we conducted a detailed code review and inspection with the primary objective of manually inspecting the smart contract code to identify any logical errors or vulnerabilities that could compromise security or functionality. This involved a comprehensive reading of each line by multiple team members, peer reviews to ensure findings' accuracy, and detailed documentation of potential issues. We also targeted common vulnerabilities specific to UPLC and the broader Cardano ecosystem, using a checklist of known issues and scanning for risky patterns associated with past security breaches. Business logic verification was critical, involving mapping out the logic flow of the contract and comparing it against intended business logic, checking for state consistency to prevent undesirable behaviors, and simulating different user roles to verify correct access controls and permissions. Additionally, the audit assessed adherence to security best practices to further ensure the robustness and security of the contract.

1.b.a - Files Audited

Below is a list of all files audited in this report, any files **not** listed here were **not** audited. The final state of the files for the purposes of this report is considered to be commit `9ec41e7619f5ba9d3dd46dd194e2146098093721` for cardano-swaps, and `1890f6f1ea74599615b0dea5829fffaed6ea9200` for cardano-loans.

Filename
cardano_swaps/lib/cardano_swaps/common/utls.ak
cardano_swaps/lib/cardano_swaps/one_way_swap/utls.ak
cardano_swaps/lib/cardano_swaps/two_way_swap/utls.ak
cardano_swaps/validators/one_way_swap.ak
cardano_swaps/validators/two_way_swap.ak
cardano-loans/aiken/validators/cardano_loans.ak
cardano-loans/aiken/lib/cardano_loans/utls.ak
cardano-loans/aiken/lib/cardano_loans/interest_validation.ak
cardano-loans/aiken/lib/cardano_loans/offer_validation.ak
cardano-loans/aiken/lib/cardano_loans/payment_validation.ak
cardano-loans/aiken/lib/cardano_loans/ask_validation.ak
cardano-loans/aiken/lib/cardano_loans/burn_and_claim_validation.ak
cardano-loans/aiken/lib/cardano_loans/burn_and_unlock_validation.ak
cardano-loans/aiken/lib/cardano_loans/accept_validation.ak
cardano-loans/aiken/lib/cardano_loans/address_update_validation.ak
cardano-loans/aiken/lib/cardano_loans/data/datums.ak
cardano-loans/aiken/lib/cardano_loans/data/redeemers.ak
cardano-loans/aiken/lib/cardano_loans/data/core.ak

2 - Findings

No critical findings were made, discussions were had with the DApp creator about optimizations - but given that there are ongoing changes to UPLC and likely a new version of the p2p DApp suite, optimizations and informational findings were excluded from this audit report. Based on our audit process, it appears that the p2p cardano-loans and cardano-swaps function as intended by the documentation with sound business logic.

3 - Appendix

3.a - Disclaimers

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Cypher Enterprises has produced this document as a general overview based on the due diligence performed on the client's smart contract. It does not guarantee the security or functionality of the smart contract either during deployment or thereafter. **CYPHER ENTERPRISES HAS NO OBLIGATION TO MONITOR THE CLIENT'S OPERATION OF THE PROJECT NOR TO UPDATE THIS DOCUMENT ACCORDINGLY.**

The content of this document may not encompass all potential vulnerabilities. It represents a thorough assessment designed to enhance the quality of the client's codebase. Nonetheless, due to the inherent risks associated with blockchain technology and cryptographic assets, including unforeseen risks and defects, complete security cannot be guaranteed.

Cypher Enterprises advises the client to conduct multiple independent audits, establish a public bug bounty program, and engage in ongoing security evaluations and monitoring. Misjudgments during the manual review process can occur, and it is prudent to obtain several independent assessments regarding critical issues. **CYPHER ENTERPRISES EMPHASIZES THAT EACH COMPANY AND INDIVIDUAL SHOULD PERFORM THEIR OWN DUE DILIGENCE AND MAINTAIN ONGOING SECURITY MEASURES.**

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Specific Disclaimer for Smart Contract Audits

The due diligence performed and the subsequent report provided by Cypher Enterprises does not guarantee the security or functionality of any smart contract at any point. This audit aims to identify potential vulnerabilities based on the state of the smart contract at the time of the audit only. Cypher Enterprises is not responsible for any changes or modifications made to the smart contract after the audit.

Encouragement of Additional Security Measures

It is strongly recommended that clients engage in comprehensive security practices, including but not limited to conducting multiple independent audits, establishing a public bug bounty program, and continuous security monitoring. Cypher Enterprises emphasizes that security is a continuous process and encourages clients to remain vigilant and proactive in addressing security concerns.

No Guarantee of Complete Security

Due to the complex and evolving nature of blockchain technology, an audit cannot guarantee complete security. Potential risks and unforeseen vulnerabilities may still exist despite thorough auditing.

Independent Due Diligence

Clients and users of this document are advised to conduct their own independent verifications and due diligence. Cypher Enterprises encourages a proactive approach to security and risk management, tailored to each client's specific circumstances.

3.b - Issue Severity Levels

3.b.a - Severity

Severity	Description
Critical	Critical issues identify severe vulnerabilities or bugs that lead to complete system failure, including total loss of data or funds with no available remedies.
Major	Major issues identify significant flaws that could lead to partial system failure, financial losses, or significant operational disruptions, albeit with some functionality remaining intact.
Minor	Minor issues point out smaller bugs or flaws that can cause inconvenience or limited disruptions but do not significantly affect the overall functionality or security of the system.
Info	Info items detail beneficial insights or observations regarding the system's operation that, while not critical, provide valuable information for optimizing performance and user experience.

3.b.b - Status

Status	Description
Resolved	Issues that have been fixed by the project team.
Acknowledged	Issues that have been acknowledged or partially fixed by the project team. Projects can decide to not fix issues for whatever reason.
Identified	Issues that have been identified by the audit team. These are waiting for a response from the project team.

3.c - About Cypher Enterprises

Cypher Enterprises is a Cardano-centric software development company that specializes in blockchain technology, smart contract development, and decentralized applications (dApps). Our team of experts is dedicated to providing high-quality services to clients in the blockchain industry, including smart contract audits, security assessments, and custom software solutions. We are committed to delivering innovative and secure solutions that meet the unique needs of our clients and contribute to the growth and success of the Cardano ecosystem.

Our engineering team has a combined experience of over 8 years in Cardano development, we have successfully been involved with the deployment of several Cardano projects. We have a deep understanding of the Cardano blockchain and its underlying technology, which allows us to provide comprehensive and reliable services to our clients.

We also operate the Summon Platform, and are working to launch the Incision Protocol. Our team is actively involved in the Cardano community and is committed to supporting the growth and development of the ecosystem. We are passionate about blockchain technology and its potential to revolutionize the way we interact with the world.